Niranjan Purushothaman

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Professional Summary

I am a Computer Science graduate experienced in frontend and backend development with a focus on building automated, intelligent solutions for real-world applications. Proficient in Full-Stack Web Development and Generative AI technologies, including LLMs, RAG, and multi-agentic architectures.

Education

Sathyabama Institute Of Science And Technology, Chennai, India **2022 – 2026**

Bachelor of Engineering (B.E.) in Computer Science and Engineering, ***CGPA: 8.05/10***

- Main subjects and concepts covered: Artificial Intelligence, Machine Learning, Web Development, Database Management Systems, Data Structures and Algorithms, Operating Systems, Computer Networks, Software Engineering.

Work Experience

Tectra Technologies, Chennai, India **Nov 2025 – May 2026**

Artificial Intelligence Intern (6 Months)

- Integrated Large Language Models (LLMs) into a CRM system to automate student interactions using a chatbot, OCR marksheet extraction tool, course recommendation system, and an assistant for maternal services offering doula and nanny booking facilities.
- Developed AI-powered voice and document intelligence pipelines using speech-to-text (WhisperX), NLP, LLMs, Python, and data extraction tools to convert unstructured audio data into structured CRM records for customer service.
- Built a full-fledged hospital chatbot for booking doctors and medical tests, answering health-related queries, and surfacing insights about hospital services.

Technical Skills

- **Programming Languages:** Python, Java, SQL
- **AI Technologies:** API Integration, AI Agents, LLMs, Multi-Agentic Systems, RAG, WhisperX, Hugging Face, AI Frameworks, NLP, Computer Vision
- **Web Development:** HTML, CSS, JavaScript, React, Node.js, FastAPI, WebSockets, REST API
- **Databases:** MySQL, MongoDB, PostgreSQL, Vector Databases, Redis
- **Tools:** Git, Docker, Postman, VS Code, Kubernetes, Cursor IDE, Antigravity, Claude Code, Android Studio

Projects

AI Integration in CRM for Student Intelligence and Automation **2026**

- Enhanced a company CRM system by integrating AI-powered features, including conversational chatbots, voice intelligence, and automated data extraction, improving user experience and significantly reducing manual workload.
- Integrated AI features into the CRM using WhisperX, OCR, LLMs, and FastAPI to automatically extract student insights from call recordings and marksheets, and developed a context-aware conversational chatbot for automated student support and intelligent data retrieval.

AI-Based SDLC Assistant with Multi-Agent Architecture **2026**

- Developed an AI-powered assistant using a multi-agent architecture to automate the requirements gathering, planning, and system design phases of the Software Development Life Cycle (SDLC), reducing manual effort by 70%.
- Implemented a Manager Agent to coordinate and monitor the Requirement, Planning, and Design agents, validate generated outputs, detect inconsistencies, and trigger regeneration for accurate results.
- Built specialized AI agents using Ollama (LLM), LangChain, LangGraph, and FastAPI to analyze user inputs and automatically generate Software Requirements Specifications (SRS), project plans, and system design documentation.

RAG-Based Conversational Chat Assistant **2025**

- Developed a document-based conversational AI chatbot using LangChain, Azure OpenAI, DeepLake vector database, and Streamlit, enabling users to upload PDFs and interact with them using a Retrieval-Augmented Generation (RAG) framework.
- Implemented vector embeddings and semantic search to retrieve relevant document information and generate accurate, context-aware responses through an interactive web interface.

Research And Publications

AI-Based SDLC Assistant with Multi-Agent Architecture **2026**

IEEE International Conference on Communication, Computing and Electronics Systems (ICCCES)

- Published a research paper in an IEEE conference (indexed in IEEE Xplore) based on my final year project, presenting a multi-agent AI system for automating Software Development Life Cycle (SDLC) phases, demonstrating the application of large language models and intelligent agents to real-world software engineering automation.
- Designed specialized AI agents, including a Requirement Agent, Planning Agent, Design Agent, and Manager Agent, to collaboratively generate Software Requirement Specifications (SRS), project plans, and system architecture.

Publication Link

Certifications

- Introduction to Generative AI - Google Cloud
- Machine Learning - Dimensionality Reduction - IBM
- Fundamentals of Encryption And Quantum Safe Techniques - IBM

Hobbies and Carrier Interests

- Chess – Improves logical thinking and concentration.
- Strength Training (Gym) – Helps maintain fitness and discipline.
- Freelancing - Developed Multiple websites and projects for clients and students.
- Currently developing an application for a Problem statement that deals with Improper rent maintenance in rented rooms.

Languages

Tamil | English (Fluent) | Telugu